Glossary: plain-language definitions

Keep this page open beside the workbook. Words are in the order you meet them. Each has a picture from everyday life.

Word	Plain meaning	Everyday picture / where you meet it
test	A tiny program that checks one thing about another program and answers pass or fail.	Tasting the soup before serving. Workbook A1.
unit test	A test of one small piece, usually one function.	Pressing the smoke-alarm button, not burning the house down. A1.
pytest	The tool that finds your tests, runs them, and reports.	The referee. A2.
assert	A line in a test that says "I claim this is true." If it is false, the test fails.	`assert length_label(179) == "short"` Step 1.
red	At least one test failed. Normal, and often good: it proves the test can notice a problem.	A2.
green	Every test passed.	A2.
stub	A placeholder function that exists but does not do the real job yet (for example, it returns `None`).	A cardboard cutout of a door. Step 1.
fixture	Made-up sample data that tests reuse. (pytest uses the word more broadly; here it is only the sample tracks in `tests/fixtures.py`.)	The same props on the stage for every scene. A4.
bin (partition cell)	A group of inputs that the function treats the same way. A **partition** cuts all inputs into bins with no overlap and nothing left out.	Postage by weight bracket. Beat 1.
boundary	The values right on either side of the edge between two bins. Test both sides.	A parcel of exactly 1 lb: which bracket? Step 1.
relation	A yes/no question about two things.	"Are these two listings the same song?" "Does A come before B?" Step 3.
property	A rule that must hold for *every* item or pair, not just one example.	"A track is a duplicate of itself." Steps 3, 6, 10, 12.
reflexive	Every item is related to itself.	Everyone sits at their own table.
symmetric	If a is related to b, then b is related to a.	"Is the sibling of." (Not: "is the boss of.") Step 10.
antisymmetric	If a is related to b *and* b is related to a, then a and b are the same item.	A photo finish with a tie breaks this. Step 12.
transitive	If a relates to b and b relates to c, then a relates to c.	"Sits near" in a row of chairs is *not* transitive. Step 6.
equivalence relation	A relation that is reflexive, symmetric and transitive. It sorts everything into groups that do not overlap: "counts as the same."	Same table number. "Same label." Steps 3, 7.
partial order	A relation that is reflexive, antisymmetric and transitive: "comes before or is equal to." Some pairs may not be comparable at all.	Socks before shoes; shirt before jacket. Steps 12, 16.
witness	A concrete example that shows a claim is true or false.	Three actual tracks with actual lengths. Step 6.
counterexample	A witness that *breaks* a rule that was supposed to always hold.	200 s, 204 s, 208 s with a gap of 5. Step 6.

Word	Plain meaning	Everyday picture / where you meet it
normalize	Rewrite things into one standard form so equal-looking things become equal.	Lowercase, drop the "(Remastered)". Step 7.
key	The standard form that normalizing produces. Two things are the same exactly when their keys are equal.	`("lantern hill", "slow orbit")`. Step 7.
characterization test	A test written to record that something already works. It starts green.	A photo of the room before you rearrange it. Step 3.
cluster / chain	Items joined into one group by a chain of matches (friend of a friend).	Beat 3.
depends on / needs	"X depends on Y" means X uses Y, so Y must work first.	Shoes need socks. Beat 7.
leaves first	Test the pieces that need nothing else first, then the pieces that rely on them.	Beat 7.
cycle	A loop of "needs": A needs B and B needs A. There is no first thing to test.	The key is locked inside the room. Step 18.
AI proposed / verified	A suggestion from Claude is only a *proposal* until you run it or check it by hand. That check makes it *verified*.	The honesty log, C2.